Super-Resolution

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EE 392J – Digital Video Processing
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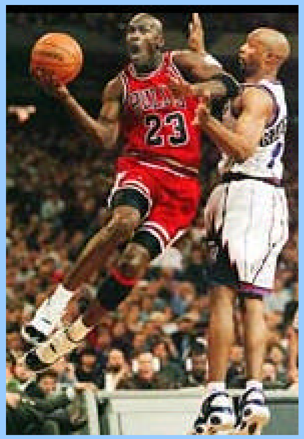
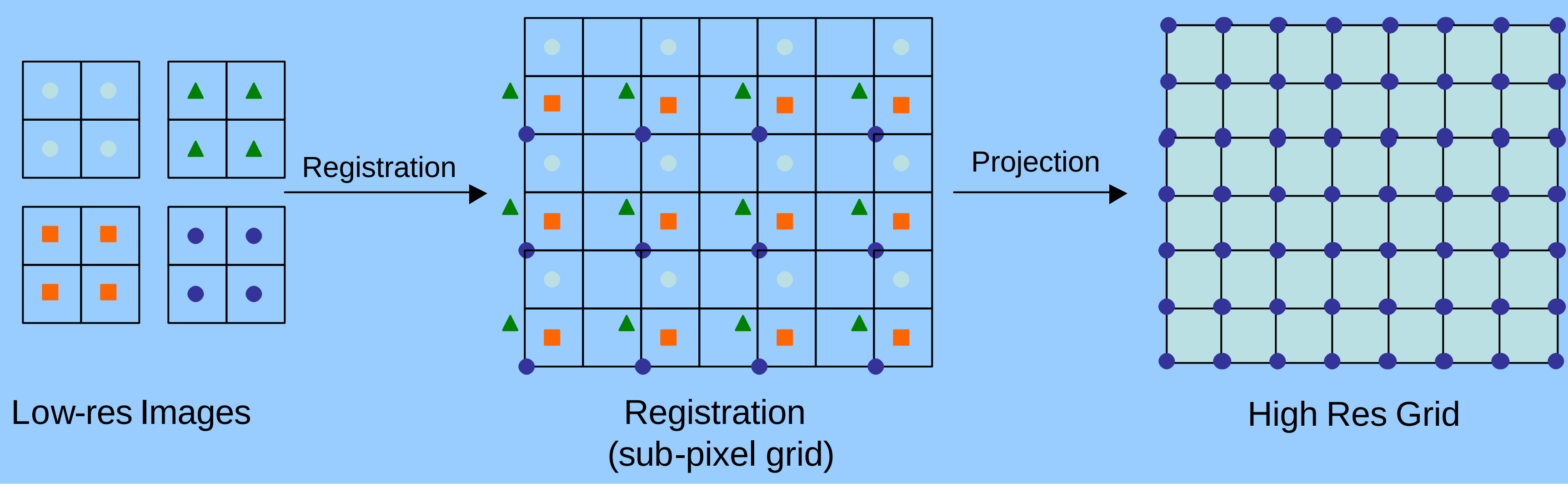
Motivation

- Create High Resolution Video from a low-resolution one
- Create High Resolution Image(s) from a video or collection of low-res images. Applications:
  - Action Packed Sports Images (Basketball dunk, Gymnastics, etc)
  - Astronomy
  - Medical Imaging
- **This project** – Create a high-res image from bunch of low-res ones (constraints: global motion – shift & rotation)



Approach

- Image Registration – Motion Estimation
- Projection onto High-Res grid
  - Nonuniform Interpolation
  - **Frequency Domain**
  - Iterative Back Projection (IBP)
  - **POCS (Projection onto convex sets)**



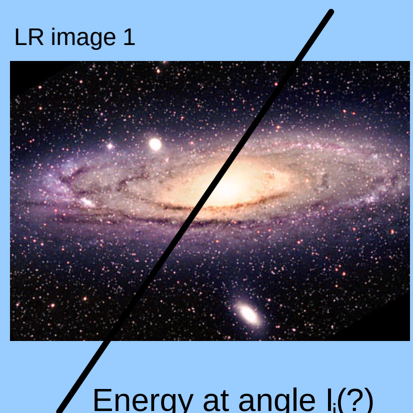
1.1 Registration (angle)

- Rotation Calculation
  - **Correlate 1st LR image with all LR images at all angles**

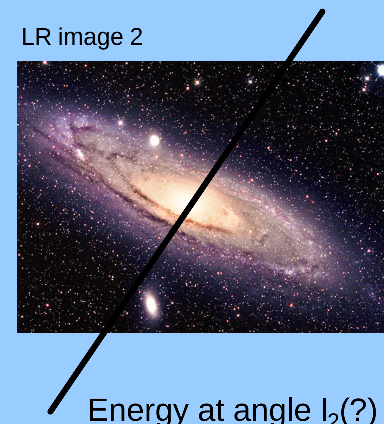
OR

  - Calculate energy at all angles for all LR images. Correlate energy vector to find the rotation angle

$$\text{Angle}_i = \text{max index}(\text{correlation}(I_1(?), I_i(?)))$$



i = 2,3,..,N (number of LR images)



1.2 Registration (shift)

- Shift Calculated using Frequency Domain Method

$$F_i(\mathbf{u}^T) = e^{j2pu?s}F_1(\mathbf{u}^T)$$

$$?\mathbf{s} = \frac{angle(\ F_i\ (\mathbf{u}^T)\ /\ F_1(\mathbf{u}^T)\ )}{2p\mathbf{u}}$$

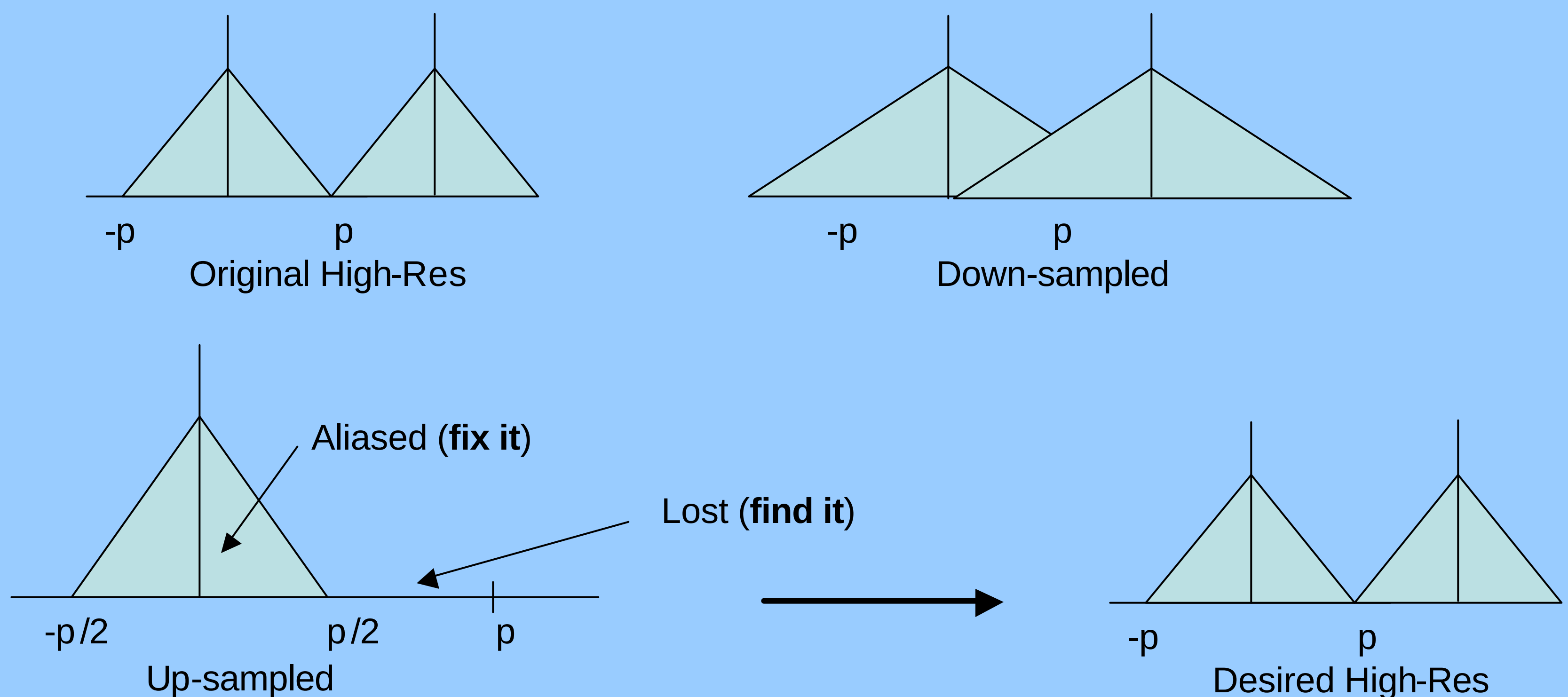
$?\mathbf{s} \rightarrow [?x\ ?y]^T$

$\mathbf{u} \rightarrow [f_x\ f_y]$

- Used only 6% lower **u** (high freq could be aliased)
- Used least square to calculate ?s

2.1 Frequency Domain

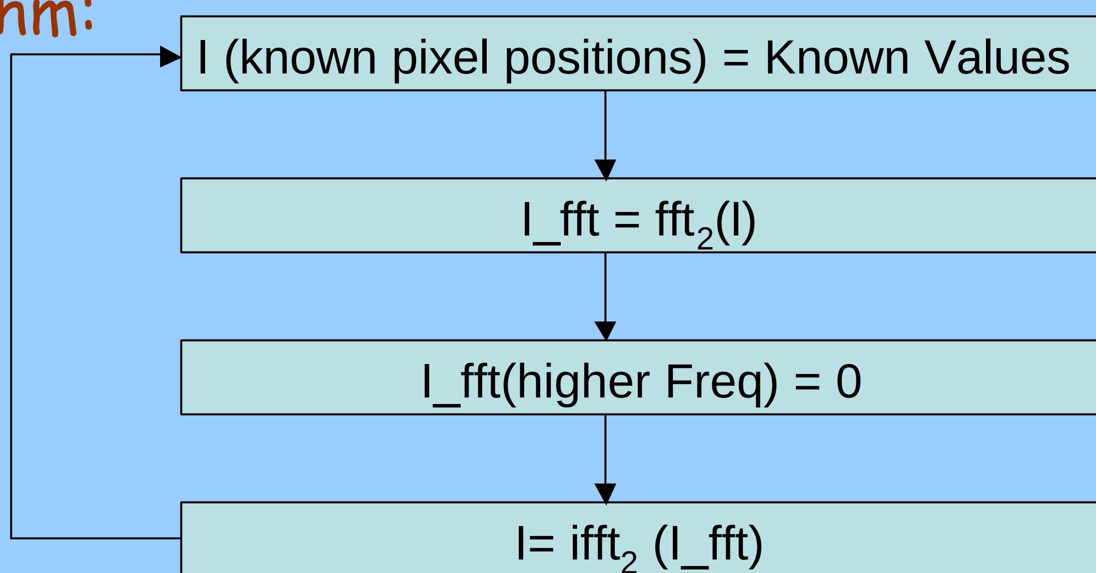
- Input → Down-sampled aliased images
- Goal I→ Correct the low-freq aliased data
- Goal II → Predict the lost high freq values



2.2 Projection onto High-res grid

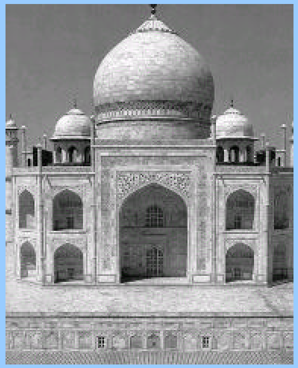
Papoulis-Gerchberg Algorithm (special case of POCS)

- Correct the low-freq values. Assumes high-freq part to be zero.
- Projection onto 2 convex sets
  - Known pixel values
  - Known Cut-off freq in the HR image
- Algorithm:

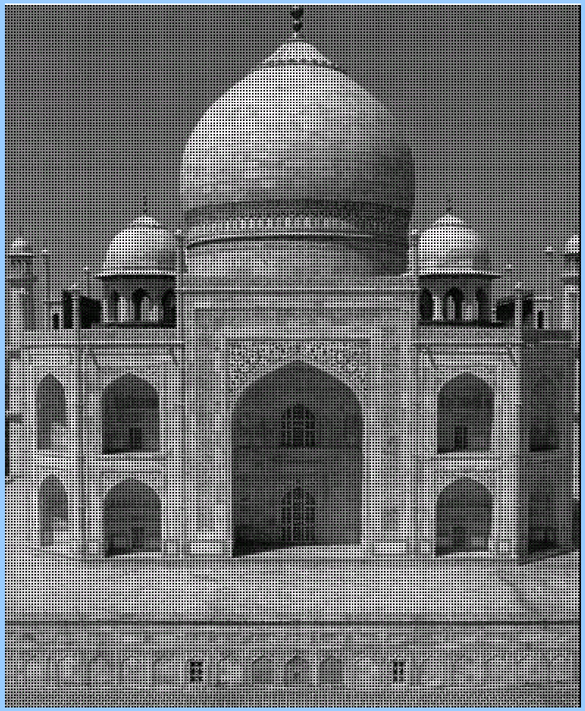


Papoulis – Gerchberg Algorithm

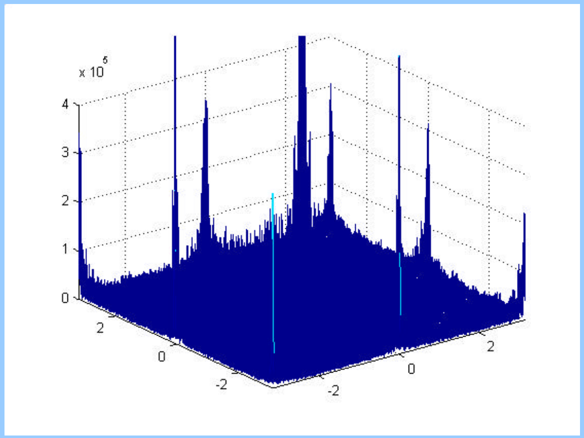
Initial Setup



Taj Mahal – Low-res image I



Reconstructed image from known pixels



FFT(Reconstructed image)

Papoulis - Gerchberg Algorithm

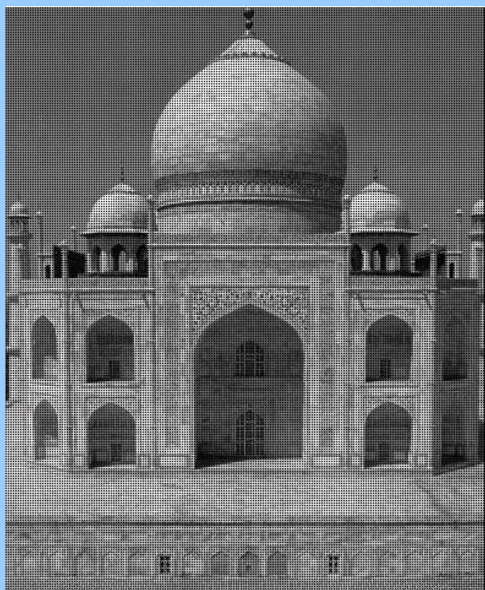
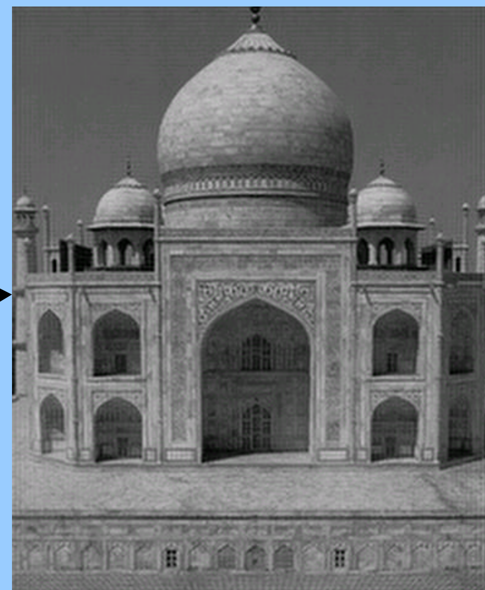


Image at iteration 0



Known Pixel Values

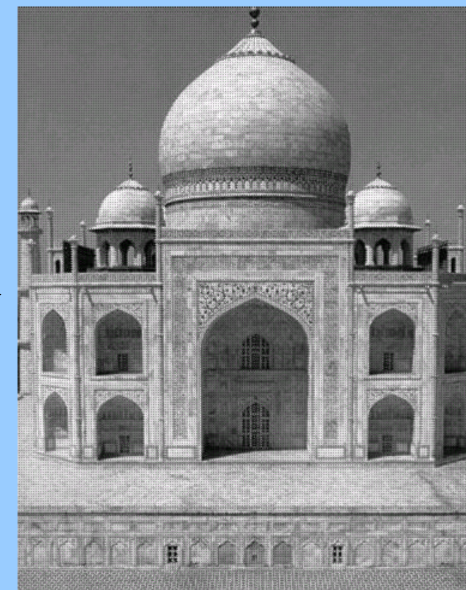
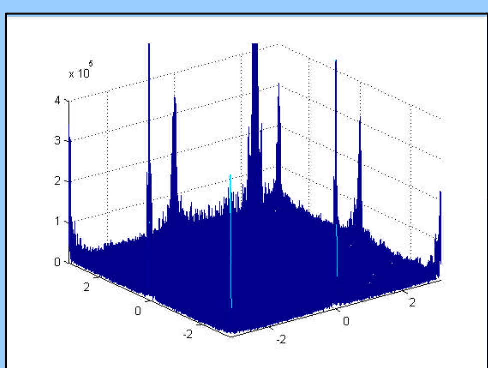
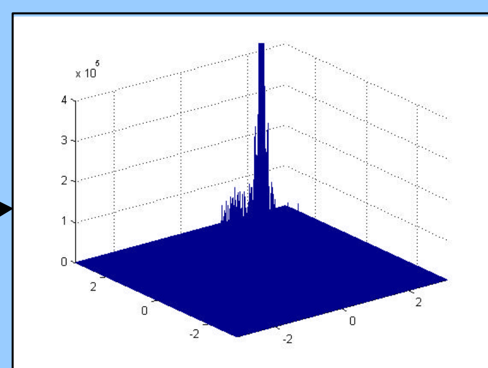
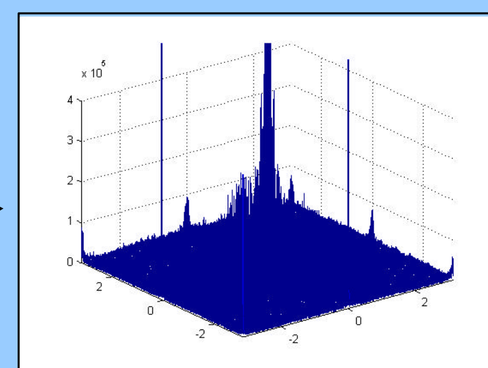


Image after 1st iteration



I(high freq) =0





FFT

Papoulis - Gerchberg Algorithm

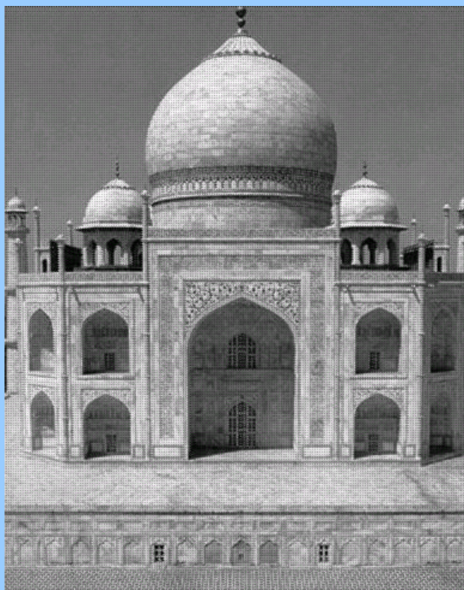
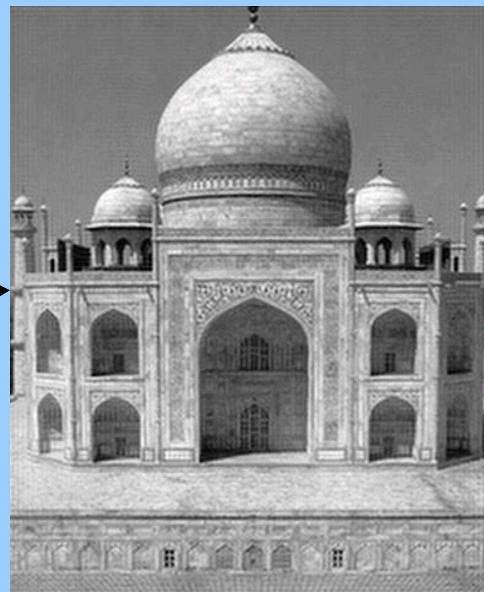


Image at iteration 1



Known Pixel Values

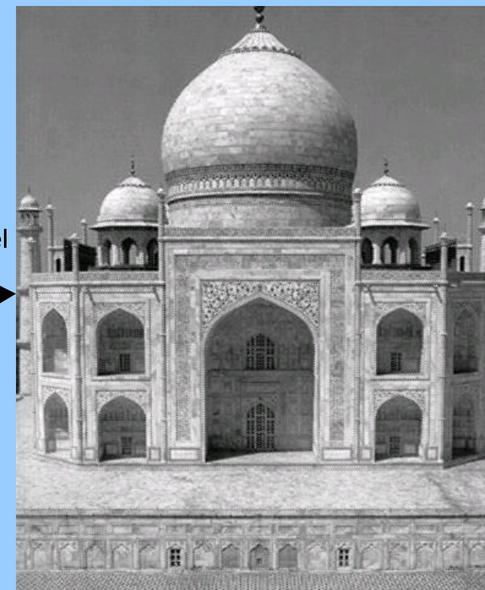
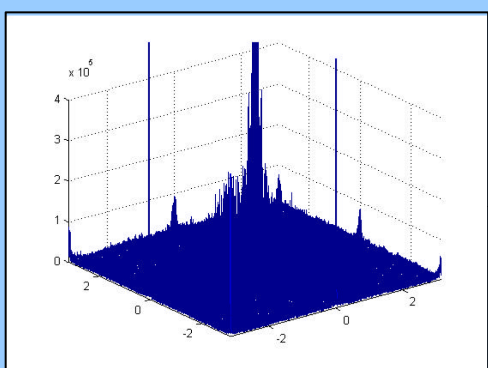
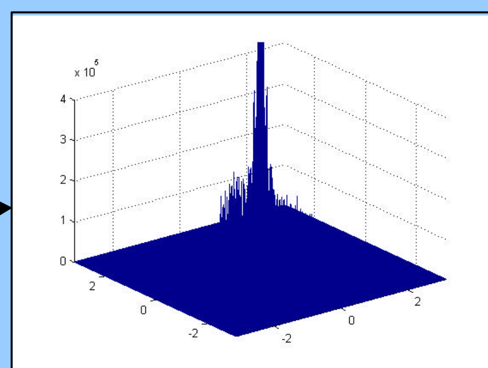
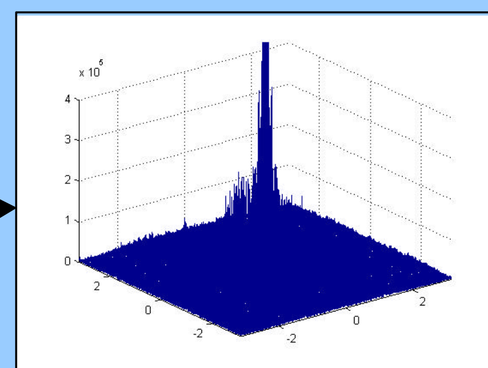


Image after 10 iterations



I(high freq) =0

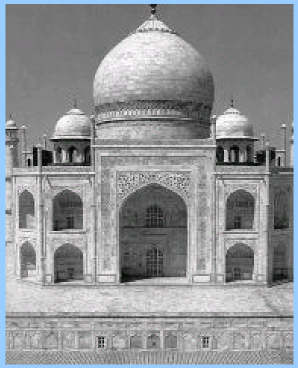




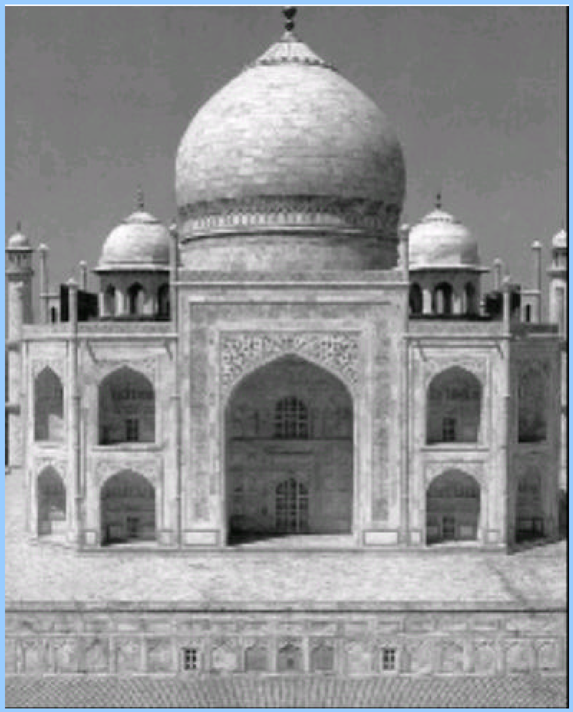
FFT

Papoulis - Gerchberg Algorithm

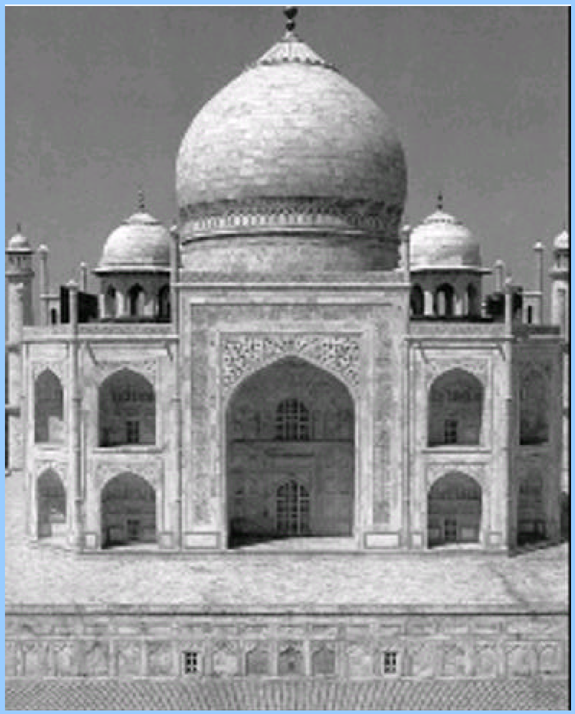
After 50 iterations



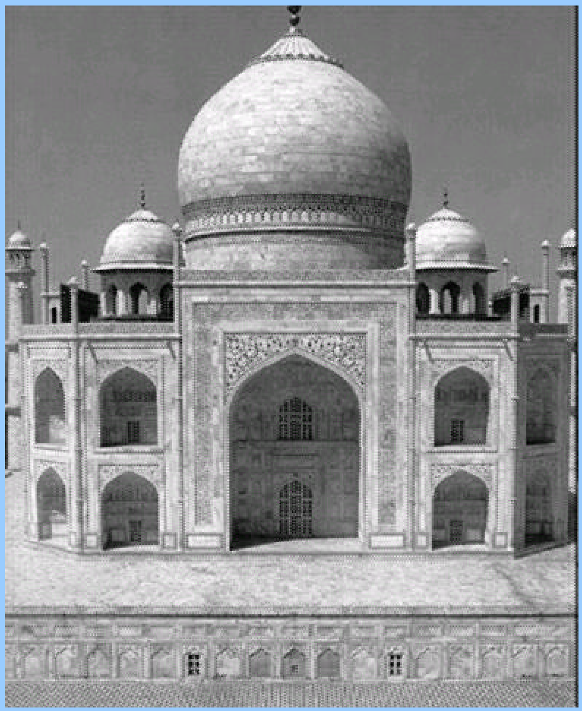
Taj Mahal – Low-res image 1



Bilinear Interpolation



Bicubic Interpolation



SR Reconstructed image

Results (Real images)

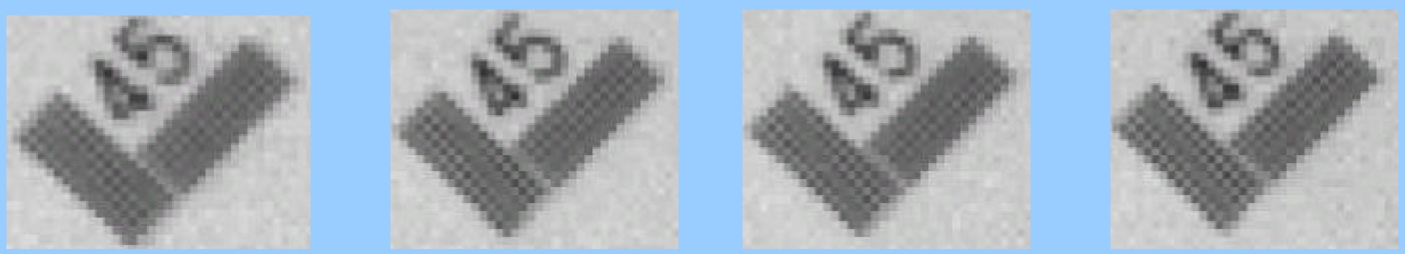
- Took 4 snaps using a high-res digital camera
- Cropped the same part of each image
- Applied SR algorithm & compared it with bicubic interpolation

Results (Synthetic Images)

- Constructed 4 low-res images by shifting and down-sampling 1 high-res image.
- Applied SR algorithm & compared it with bicubic interpolation

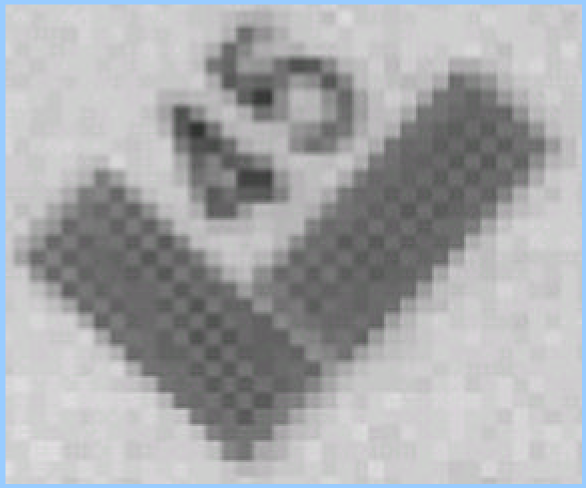
Results (Real Images - I)

Original Low-res images
(Courtesy: Patrick Vandewalle)



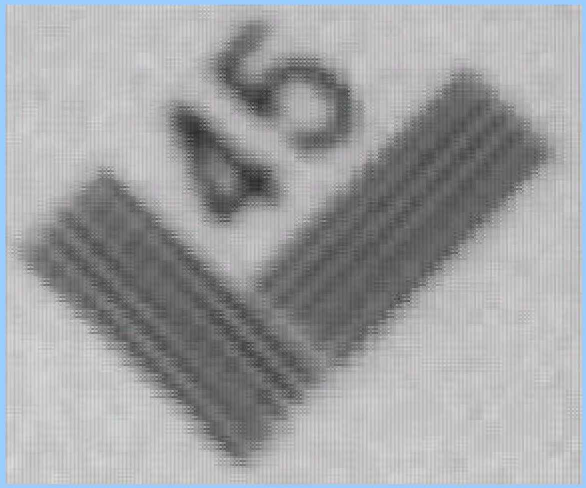
Results (Real Images - I)

Bicubic Interpolation



Results (Real Images - I)

Super-resolution



Results (Real Images - II)

Low-Res Image I

Low-Res Image II

- Didn't WORK !!!
- Motion was not restricted to shifts & rotation
- Images had affine mapping.
- **Rule I – Need Correct Registration**

Results (Synthetic Image - I)

Original High-Res

Down-sampled

Results (Synthetic Image - I)

Bicubic Interpolation

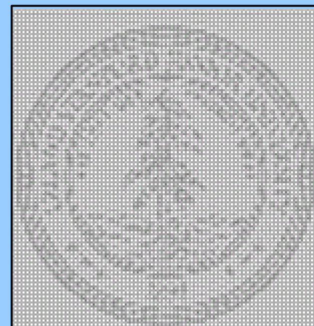
Results (Synthetic Image - I)

Super-Resolution

Results (Synthetic Image - II)

Original

Bicubic

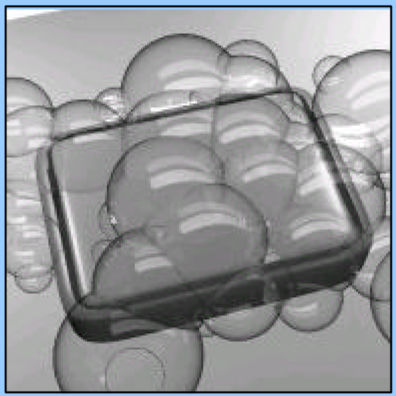


SR

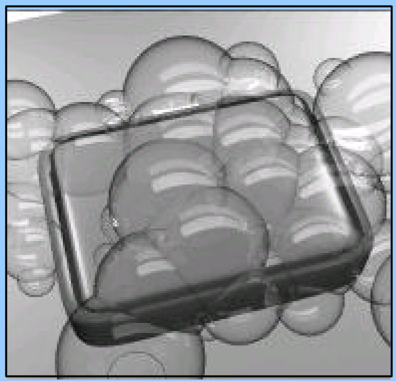
Why didn't SR work???

- Low-res images were created by forcing shifts at critical velocities
- **Rule II → If low-res images are at critical velocities, can't create good HR image**

Results (Synthetic Image - III)


Original


Bicubic

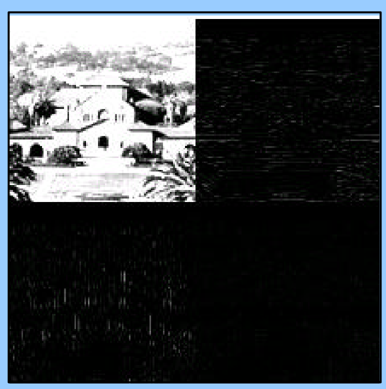

SR

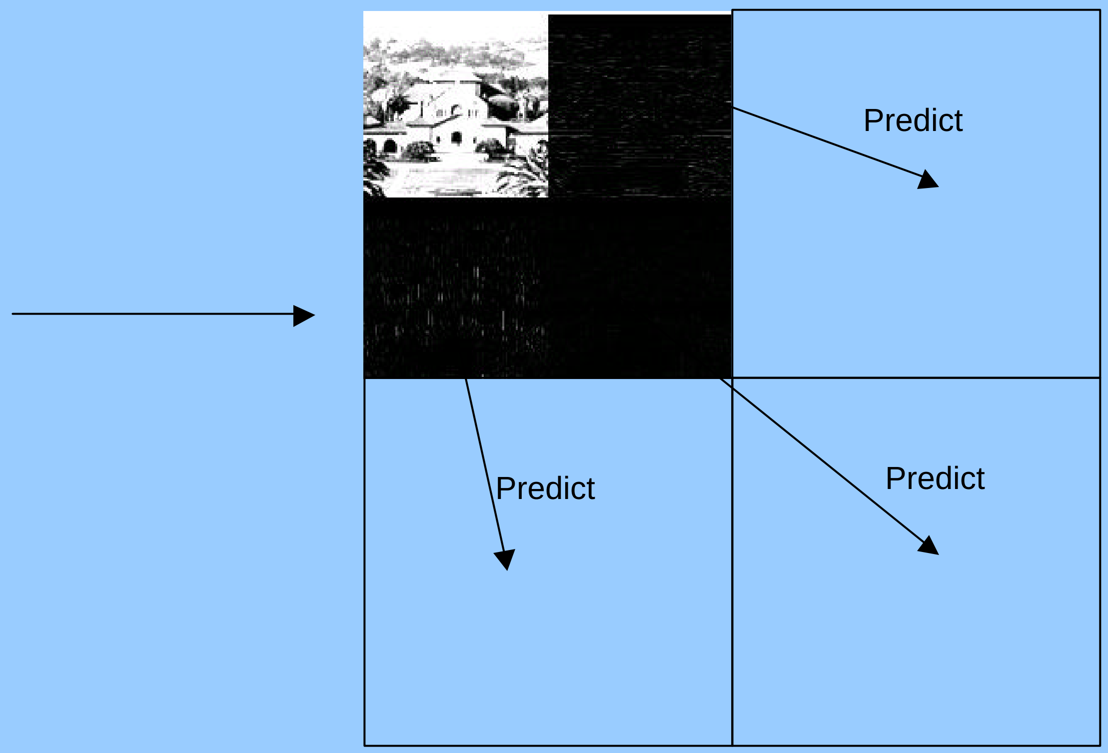
Why did SR work so well???

- Low-res images were created by forcing shifts at non-critical velocities
- **Rule III → If low-res images have all the info about high-res then HR image can be perfectly constructed**

Future Work

- Superresolution with multiple motions between frames → create high res video
- Predict the high-res frequency components using wavelet methods





Acknowledgements

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- Prof Susie Wee
- Patrick Vandewalle

- Q & A ???
- Comments !!!!